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SUBJECT :ETHICAL HACKING

SUB.CODE :ITA1407

CO3 -ASSESSMENT 3

The security issue is that the script stores **passwords in plaintext**, meaning anyone who gains access to the script, database, configuration file, logs, or backup may be able to read the passwords directly.

Security Risks

- **Credential theft:** Attackers can immediately obtain users' passwords.
- **Account compromise:** Users who reuse passwords on other systems may have additional accounts compromised.
- **Insider threat:** Employees with access to the storage location could view credentials.
- **Data breach impact:** A compromised database exposes usable credentials rather than protected password representations.

- **Compliance issues:** Plaintext password storage violates common security best practices and may conflict with applicable security/privacy requirements.

Recommended Solution

Passwords should **never be stored in plaintext**. For authentication systems, store passwords using a dedicated password-hashing algorithm such as:

- **Argon2id** — preferred modern option.
- **bcrypt**
- **scrypt**

Use a unique salt for every password and rely on a well-tested library rather than implementing the hashing algorithm manually.

Conceptually:

User password



Password hashing algorithm + unique salt



Stored password hash

During login, the submitted password is hashed/verified against the stored hash rather than compared with a plaintext password.

Other sensitive information should also be protected using appropriate **encryption, access controls, secrets management, and secure configuration practices.**